FINC5001

Module 5

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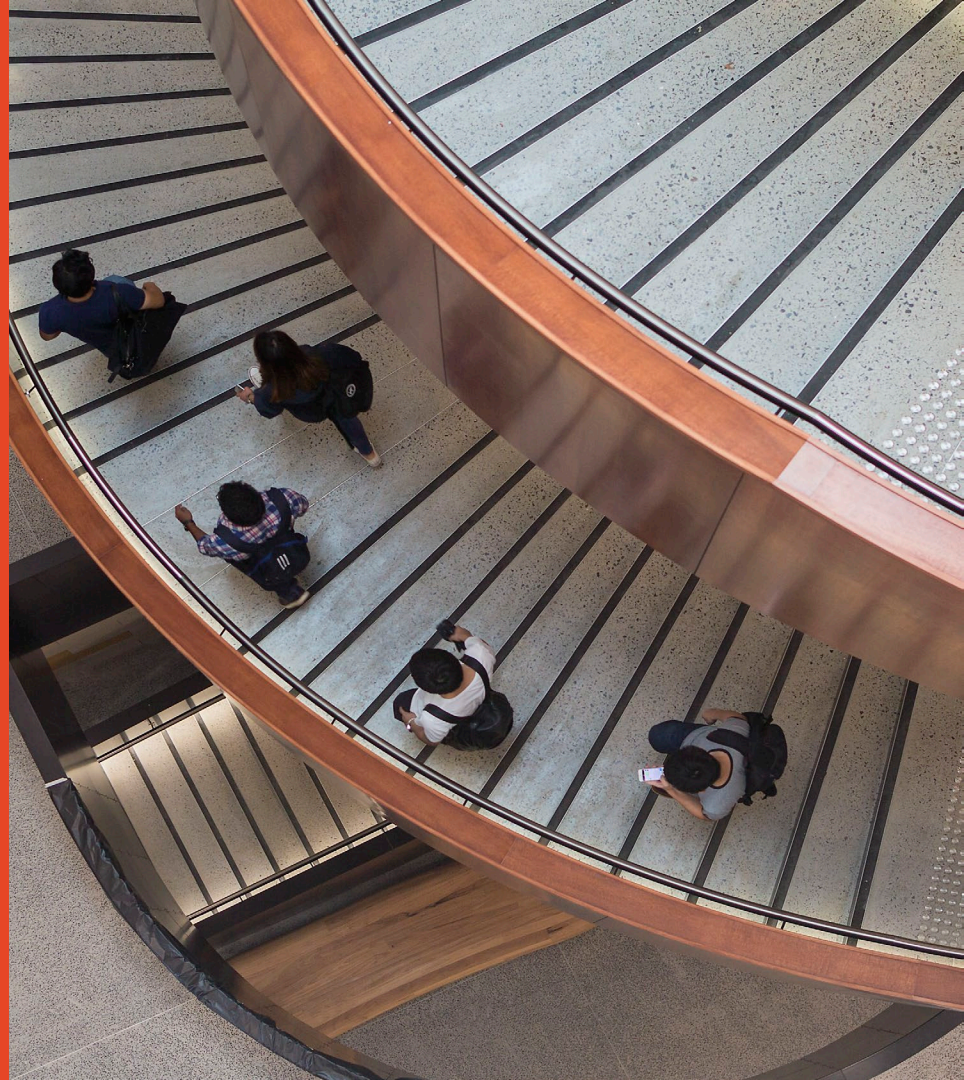


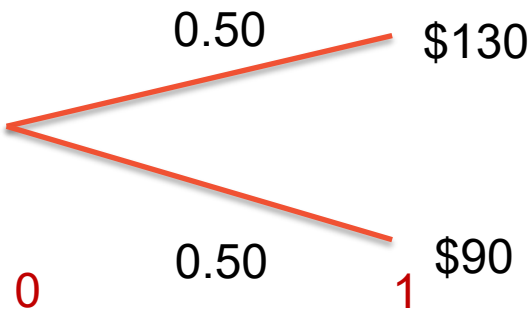
Assume the following opportunity:

- Suppose we have an asset with a certain payoff of $110 in one-years time

- What will be the asset's price if the return on another risk-free asset is 10%?
- $P_0 = \$110/1.10 = \$100$
- But what if the future payoff was risky?

How would you price the following?

- Suppose we have an asset with a 50% chance of returning $130 and a 50% chance of returning $90 in one years time



- The expected cash flow from the asset would be $110 (ie $130 x 0.5 plus $90 x 0.5)
- Would you still be prepared to pay $100 for this asset?
- Higher uncertainty requires a higher return

Interpretation of Risky Discount Rates

- *Hence:*
- risky required return = risk free rate% + risk premium%

- *note also that:*
- risk premium% = risky required return% - risk free rate%

- *This lecture and the next are devoted to building a model of the risk premium!*

Learning objectives

- Calculate realised and expected rates of return and risk
- Describe the historical pattern of financial market returns
- Compute geometric (or compound) and arithmetic average rates of return
- Explain how investors make decisions and what their preferences are
- Determine the expected return and variance of a portfolio with 2 and 3 assets
- Discuss the concept of diversification and its limits
- Discuss concepts that are fundamental to understanding CAPM (coming in the next module)

Calculating the realised return from an investment

- **Realised return or cash return** measures the gain or loss on an investment.
- *Example*: You invested in 1 share of Apple (AAPL) for $95 and sold a year later for $200. The company did not pay any dividend during that period. What will be the cash return on this investment?

$$\text{Cash return} = \text{Ending price} + \text{Cash distribution (dividend)} - \text{Beginning price}$$

- Cash return $= \$200 + 0 - \$95$
  $= \mathbf{\$105}$

Calculating the realised return from an investment (cont.)

- We can also calculate the rate of return as a *percentage*. It is simply the cash return divided by the beginning stock price

$$\text{Rate of return} = \frac{\text{Cash return}}{\text{Beginning price}} = \frac{\text{Ending price} + \text{Cash distribution (dividend)} - \text{Beginning price}}{\text{Beginning price}}$$

- *Example*: Calculate the rate of return for the previous example

- Rate of return = (\$200 + 0 – \$95) ÷ 95
  = **110.53%**

Measuring an investor's realised rate of return from investing in ordinary shares

	Share Prices		Cash distribution (dividend)	Return	
	Beginning (1 Aug 2013)	Ending (1 Aug 2014)		Cash	Rate
Company	A	B	C	D = C + B − A	E =D/A
Ten Network Holdings Ltd (TEN)	$ 0.27	$ 0.29	$ 0.00	$ 0.02	7.4%
Harvey Norman Holdings Ltd (HVN)	2.76	3.06	0.105	0.405	14.7%
Flight Centre Travel Group Ltd (FLT)	45.94	47.00	1.46	2.52	5.5%
Origin Energy Ltd (ORG)	12.16	14.23	0.50	2.57	21.1%
Coca-Cola Amatil Ltd (CCL)	12.90	9.25	0.56	(3.09)	−24.0%

Legend:
We formalise the return calculations found in columns D and E using equations (7–1) and (7–2):

Column D (cash or dollar return)

$$\text{Cash return} = \text{Ending price} + \text{Cash distribution (dividend)} - \text{Beginning price} = P_{End} + Div - P_{Beginning} \quad (7\text{–}1)$$

Column E (rate of return)

$$\text{Rate of return } (r) = \frac{\text{Cash return}}{\text{Beginning price}} = \frac{P_{End} + Div - P_{Beginning}}{P_{Beginning}} \quad (7\text{–}2)$$

Source: Share price and cash distribution (dividend) information from Yahoo! Finance, <http://finance.yahoo.com>.

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Calculating the expected return from an investment

- **Expected return** is what the investor expects to earn from an investment in the future.

- It is the weighted average of the possible returns, where the weights are determined by the probability that it occurs.

$$\begin{array}{c}\text{Expected}\\ \text{rate of return}\\ [E(r)]\end{array} = \begin{pmatrix}\text{Rate of}\\ \text{return 1}\\ (r_1)\end{pmatrix} \times \begin{pmatrix}\text{Probability}\\ \text{of return 1}\\ \Pr(r_1)\end{pmatrix} + \begin{pmatrix}\text{Rate of}\\ \text{return 2}\\ (r_2)\end{pmatrix} \times \begin{pmatrix}\text{Probability}\\ \text{of return 2}\\ \Pr(r_2)\end{pmatrix} + \cdots + \begin{pmatrix}\text{Rate of}\\ \text{return } n\\ (r_n)\end{pmatrix} \times \begin{pmatrix}\text{Probabilty}\\ \text{of return } n\\ \Pr(r_n)\end{pmatrix}$$

Calculating the expected rate of return for an investment in ordinary shares

State of the economy	Probability of the state of the economy*	End-of-year selling price for the shares	Beginning price of the shares	Cash return from your investment	Percentage rate of return = Cash return/Beginning price of the share	Product = Rate of return × Probability of state of the economy
Column A	Column B	Column C	Column D	Column E = C − D	Column F = E ÷ D	Column G = B × F
Recession	20%	$ 9000	$10000	$(1000)	−10% = −$1000 ÷ $10000	−2.0%
Moderate growth	30%	11 200	10000	1200	12% = $1200 ÷ $10000	3.6%
Strong growth	50%	12 200	10000	2200	22% = $2200 ÷ $10000	11%
Sum	100%					12.6%

*The probabilities assigned to the three possible economic conditions have to be determined subjectively, which requires management to have a thorough understanding of both the investment cash flows and the general economy.

Calculating the expected return from an investment (cont.)

– **Example**

– Expected return $= (-10\% \times 0.2) + (12\% \times 0.3) + (22\% \times 0.5)$

$= \mathbf{12.6\%}$

Measuring risk (cont.)

- **Variance** is the average squared difference between the individual realised returns and the expected return.

- **Standard deviation** is the square root of the variance and is more commonly used to quantify risk.

Calculating the variance and standard deviation of the rate of return

- Assume two possible investment alternatives:
  1. Australian Treasury bond – This Treasury bond is considered **risk-free** as there is no risk of default on the promised payments of 5%.
  2. Ordinary shares of the Ace Publishing Company – An investment in common stock will be a risky investment

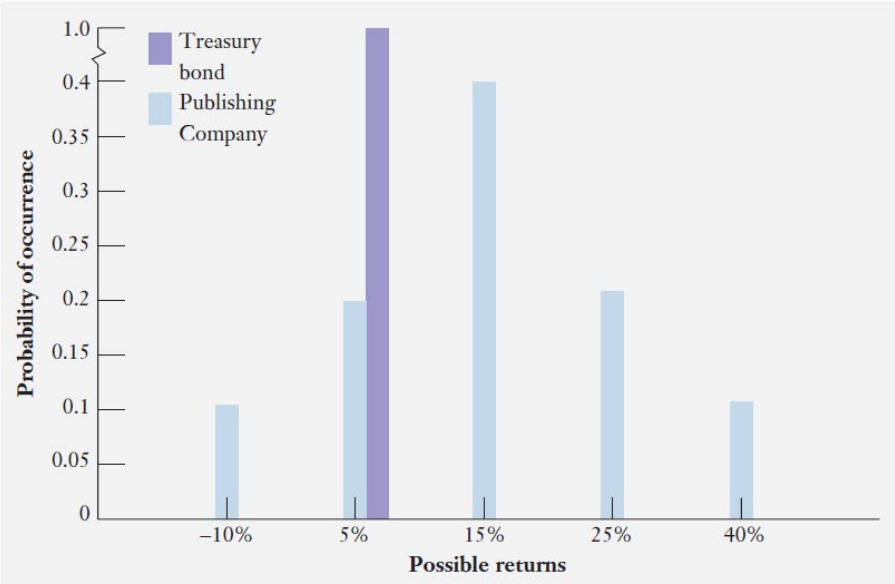
Calculating the variance and standard deviation of the rate of return (cont.)

- The **probability distribution** of an investment's return contains all possible rates of return from the investment along with the associated probabilities for each outcome

Probability distribution of returns for a Treasury Bill and Ace Publishing Company

Probability distribution of returns for a Treasury bond and the ordinary shares of the Ace Publishing Company

A probability distribution provides a tool for describing the possible outcomes or rates of return from an investment and the associated probabilities for each possible outcome. Technically, the following probability distribution is a discrete distribution because there are only five possible returns that the Ace Publishing Company shares can earn. The Treasury bond investment offers only one possible rate of return (5%) because this investment is risk-free.



Chance or probability of occurrence	Rate of return on investment
1 chance in 10 (10%)	−10%
2 chances in 10 (20%)	5%
4 chances in 10 (40%)	15%
2 chances in 10 (20%)	25%
1 chance in 10 (10%)	40%

Calculating the variance and standard deviation of the rate of return (cont.)

- The probability distribution for the Treasury bond is a single spike at 5% rate of return indicating that there is 100% probability that you will earn 5%.

- The returns for Ace Publishing company range from a low of -10% to a high of +40%. Thus the ordinary shares investment is risky, whereas the Treasury bond is not.

Calculating the variance and standard deviation of the rate of return (cont.)

- Using these figures, expected return on the stock is 15% while the expected return on the Treasury bond is 5%.

- Does the higher return of the share make it a better investment? Not necessarily, we also need to know the risk in both the investments.

Calculating the variance and standard deviation of the rate of return (cont.)

- Risk, as measured by variance, for time-series data:

$$\sigma^2 = \frac{\sum_{t=1}^{n} (r_t - \bar{r})^2}{n-1}$$

- Where:

r_t = the observed return during interval t

$\bar{r}$ = the mean (expected) return on the security

n = the number of observations

Calculating the variance and standard deviation of the rate of return (cont.)

- Risk, as measured by variance, for probability distribution data data:

$$\sigma^2 = \sum_{t=1}^{n} (r_t - \bar{r})^2 \times p_i$$

- where:

  r_t = the observed return during interval t

  $\bar{r}$ = the mean (expected) return on the security

  p_i = the probability of outcome *i*

Measuring the variance and standard deviation of Ace Publishing's Shares

State of the world	Rate of return	Chance or probability		Step 2	Step 3
a	b	c	$d = b \times c$	$e = [b - E(r)]^2$	$f = e \times c$
1	−0.10	0.10	−0.01	0.0625	0.00625
3	0.05	0.20	0.01	0.0100	0.00200
4	0.15	0.40	0.06	0.0000	0.00000
4	0.25	0.20	0.05	0.0100	0.00200
5	0.40	0.10	0.04	0.0625	0.00625

Step 1: Expected return ($E(r)$) = → 0.15
Step 4: Variance = → 0.0165
Step 5: Standard deviation = → 0.1285

Calculating the variance and standard deviation of the rate of return on an investment (cont.)

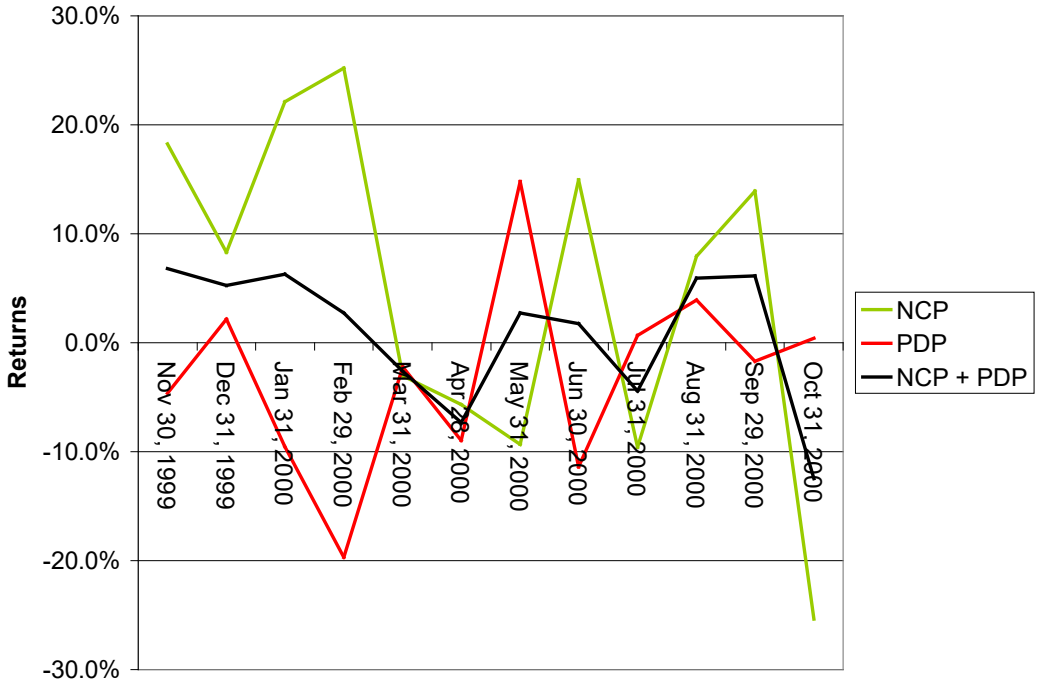
Investment	Expected return	Standard deviation
Treasury bond	5%	0%
Ordinary shares	15%	12.85%

- We observe that the shares offer a higher expected return but also entail more risk, as measured by standard deviation. An investor's choice of a specific investment will be determined by their attitude toward risk

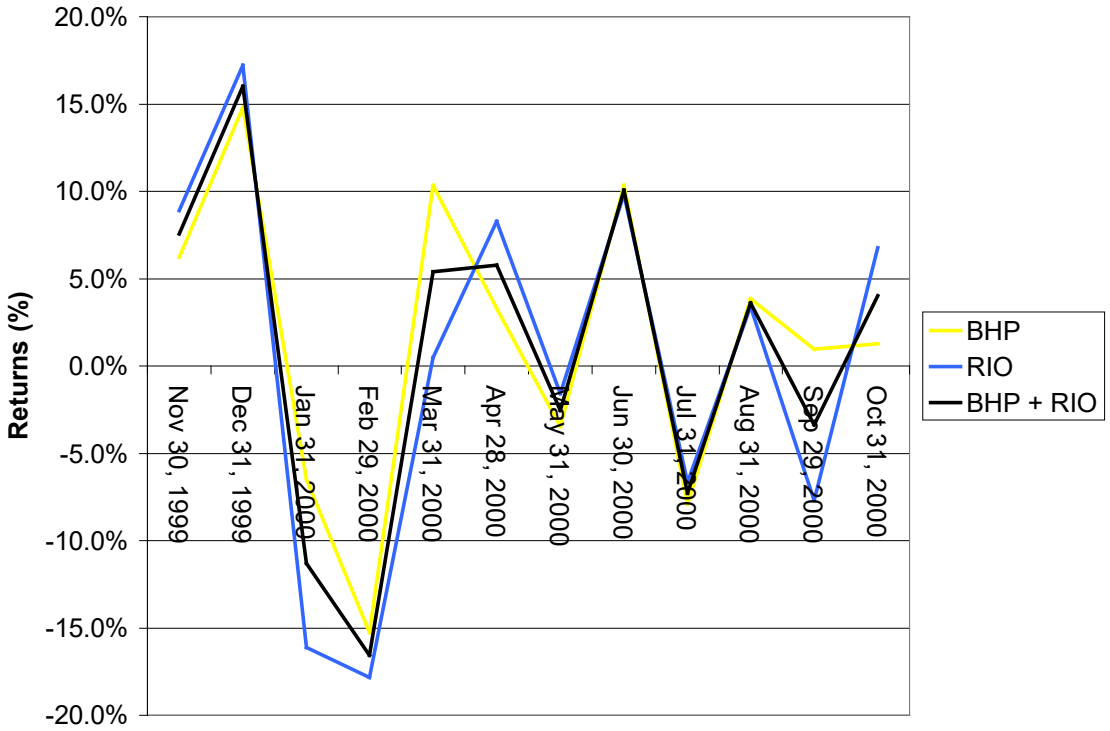
The risk of portfolios

- A ‘portfolio’ is a group of single assets held by an investor
- Holding a portfolio of stocks reduces investment risk, because:
  - Within the portfolio, individual stocks are more volatile than the overall portfolio
  - Poor returns on some stocks are offset by stronger returns on others
  - The overall portfolio return is “smoothed out”

Combining stocks in portfolios to reduce risk



Combining stocks in portfolios to reduce risk (cont)



The risk of portfolios (cont)

Example

Consider the return and standard deviation for News Corp (NWS) and Newcrest Mining (NCM), individually and combined in a portfolio, from July 2009 to July 2010

	Stock price ($)			Standard Deviation
Investment	July 2009	July 2010	Return (%)	
NWS only	14.63	16.40	12.09	7.73%
NCM only	2.73	4.31	57.88	11.59%
Portfolio	200.00	269.97	34.99	6.51%

The standard deviation of monthly returns on the portfolio was considerably lower than that of each stock individually

Covariance

– A raw measure of the degree of association between two variables is the *covariance*:

$$\sigma_{ij} = \frac{\sum_{t=1}^{n} \left(r_{i;t} - \bar{r}_i\right)\left(r_{j;t} - \bar{r}_j\right)}{n-1}$$

where:

$r_{i;t}$	=	return on security *i* during interval *t*
$r_{j;t}$	=	return on security *j* during interval *t*
r_i	=	the mean return on security *i*
r_j	=	the mean return on security *j*
n	=	the number of observations

Covariance (cont)

- If a covariance is negative:
  - An increase in returns on asset *i* is associated with a decrease in returns on asset *j*, *or*
  - A decrease in returns on asset *i* is associated with an increase in returns on asset *j*
- Hence:
  - Negative covariance → negatively correlated returns on two securities
  - Positive covariance → positively correlated returns on two securities

Covariance (cont)

Example

Calculate the covariance of monthly returns on NWS and NCM for the last four months of 2008

2008	Return on NWS	Return on NCM
Sept	0.0623	0.0110
Oct	-0.0456	0.0870
Nov	-0.0150	0.1867
Dec	0.1839	-0.0871

Label NWS security *i*, and NCM security *j*.

Calculate the mean return on NWS and NCM :

$$E(r) = \frac{\sum_{t=1}^{n} r_t}{n}$$

$$E(r_i) = (0.0623 - 0.0456 - 0.0150 + 0.1839)/4 = 0.0464 = 4.64\%$$

$$E(r_j) = (0.0110 + 0.0870 + 0.1867 - 0.0871)/4 = 0.0494 = 4.94\%$$

Covariance (cont)

Example (cont)

Calculate the covariance of monthly returns on NWS and NCM for the last four months of 2008

2008	Return on NWS	Return on NCM
Sept	0.0623	0.0110
Oct	-0.0456	0.0870
Nov	-0.0150	0.1867
Dec	0.1839	-0.0871

Calculate the covariance:

$$\sigma_{ij} = \sum_{t=1}^{n} \left(r_{i;t} - \bar{r}_i\right)\left(r_{j;t} - \bar{r}_j\right) \div (n-1)$$

$$\sigma_{ij} = \begin{bmatrix} (0.0623 - 0.0464) \times (0.0110 - 0.0494) \\ + (-0.0456 - 0.0464) \times (0.0870 - 0.0494) \\ + (-0.0150 - 0.0464) \times (0.1867 - 0.0494) \\ + (0.1839 - 0.0464) \times (-0.0871 - 0.0494) \end{bmatrix} \div 3 = -0.010$$

Correlation coefficient

– An alternative, and closely related, measure of the degree of association between two securities is the *correlation coefficient*, calculated as follows:

$$\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

where:

ρ_{ij} = the correlation coefficient between *i* and *j*

σ_{ij} = the covariance between returns on securities *i* and *j*

σ_i = the standard deviation of security *i*

σ_j = the standard deviation of security *j*

Correlation coefficient (cont)

- The correlation coefficient takes the covariance between two securities and divides (or *standardises*) it by the product of the two securities' standard deviations
- The effect of this standardisation is that the correlation coefficient will always lie between
  +1 and –1
- This provides a more interpretive measure of the strength of association between the securities – not just whether it is positive or negative

Correlation coefficient (cont)

Example

Calculate the correlation coefficient for NWS and NCM for 2009/2010, given that the standard deviation of returns on the two stocks were 7.73% and 11.59% and the covariance is –0.000656.

Label NWS security *i*, and NCM security *j*.

$$\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} = \frac{-0.000656}{0.0773 \times 0.1159} = -0.073$$

Correlation coefficient (cont)

- If $\rho = 1$ the securities are said to be *perfectly positively correlated* – returns vary in a directly uniform manner
- If $\rho = -1$ they are said to be *perfectly negatively correlated* – changes in returns are inversely proportional
- If $\rho = 0$, there is no association between returns on the two securities
- A correlation coefficient of –0.073 indicates a mild negative relationship between the return on NWS and NCM

Calculating the risk of portfolios

- The risk of a portfolio is related to the riskiness of the stocks and the degree of correlation
- The variance of returns (σ_p^2) on a portfolio is calculated as follows:

$$\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}$$

where:

σ_1, σ_2 = the standard deviation of returns on asset *1* and *2*

x_1, x_2 = proportion of the portfolio invested in asset *1* or *2*

σ_{12} = the covariance between assets *1* and *2*

Portfolio Risk

Example

Calculate the risk of a \$1 million portfolio, comprising \$0.5 million in NWS and \$0.5 million in NCM. Their standard deviations of returns are 7.73% and 11.59% and the covariance of return on the stocks is -0.000656.

Label NWS security *i*, and NCM security *j*.

$$\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}$$

$$\sigma_p^2 = \left(0.5^2 \times 0.0773^2\right) + \left(0.5^2 \times 0.1159^2\right)$$

$$+ \left(2 \times 0.5 \times 0.5 \times -0.000656\right) = 0.004524$$

$$\sigma_p = \sqrt{0.004524} \approx 0.0673 \text{ or } 6.73\%$$

Components of risk

- The risk of investing in a single security can be reduced by combining the security with others in a portfolio
- **However, this does not mean that all risk can be eliminated**
- This is because there are two components of total risk
  - Unsystematic (or diversifiable) risk
  - Systematic (or non-diversifiable) risk

Unsystematic risk

- Unsystematic risk relates to price movements that are caused by an event that influences a single company alone
- It is known as diversifiable risk because it can be diversified away
- *Example:* The BHP price fell 4% on the morning of 19 September 1997, even though the All Ordinaries was relatively stable. This was caused by a poor quarterly earnings announcement by BHP

Systematic risk

- Systematic risk relates to macroeconomic events that affect the prices of all securities, and are reflected in broad market movements
- It is referred to as 'non-diversifiable' as it is common to all securities and cannot be diversified away
- *Example:* Price movement caused by the Asian currency crisis of October 1997 had a dramatic effect on the Australian market

A well-diversified portfolio faces market risk alone, and the level of market risk will depend on the market risk of stocks included in the portfolio

Possible portfolios- two shares

- The return and risk possibilities that can be achieved by combinations of stocks are called feasible portfolios
- We will examine several feasible portfolios using BHP Billiton Ltd and Wesfarmers Ltd shares

Example

Daily risk-return characteristics for BHP and WES, 2010

	BHP	WES
Mean (%)	0.0298	0.0186
Standard deviation (%)	1.4346	1.3733
Covariance with BHP		0.000096

Possible portfolios- two shares (cont)

Example

Feasible portfolios containing BHP and WES, 2010

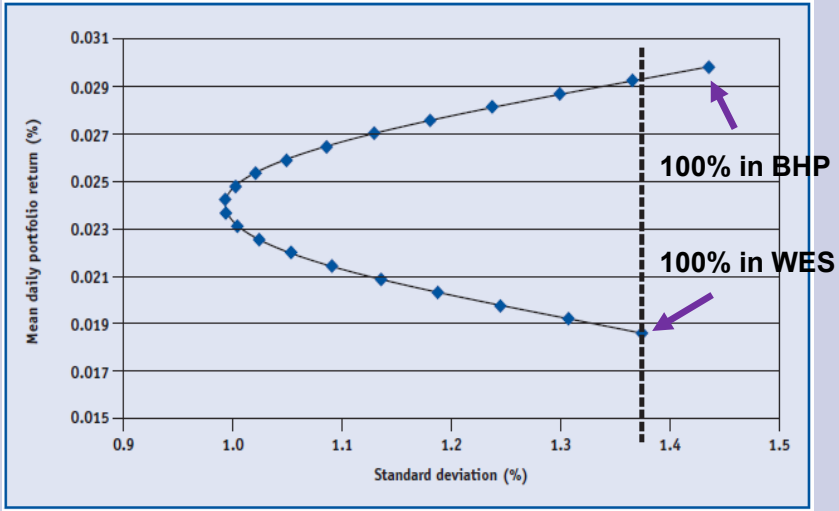
Portfolio number	Proportion in portfolio		Standard deviation	Mean return (%)
	BHP	WES		
1	1	0	1.4346	0.0298
5	0.8	0.2	1.1801	0.0275
15	0.3	0.7	1.0532	0.0220
21	0	1	1.3733	0.0186

This table shows the mean return and standard deviation for a few of the 21 feasible portfolios containing BHP & WES where the weights of each stock are multiples of 0.05

Possible portfolios- two shares (cont)

Example

Feasible portfolios containing RIO and ORI, 2007



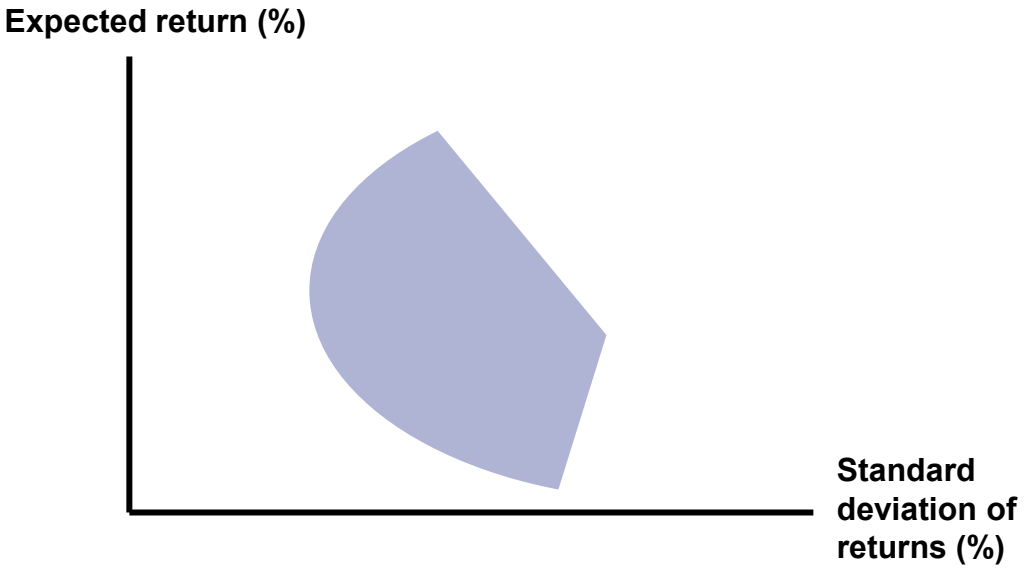
All portfolios to the left of the dotted line have less risk than either BHP or WES individually

This chart shows the risk/return characteristics for 21 different feasible portfolios of BHP & WES

Possible portfolios – multiple stocks

- It is almost impossible to calculate the risk and return of *all* feasible portfolios
- However, it has been proven mathematically that the set of feasible portfolios for a large number of stocks is enveloped by a parabola
- This set of feasible portfolios is referred to as the ***opportunity set*** of risky portfolios
- Each point within the opportunity set represents a possible level of risk and return that an investor can achieve

Possible portfolios – multiple stocks (cont)



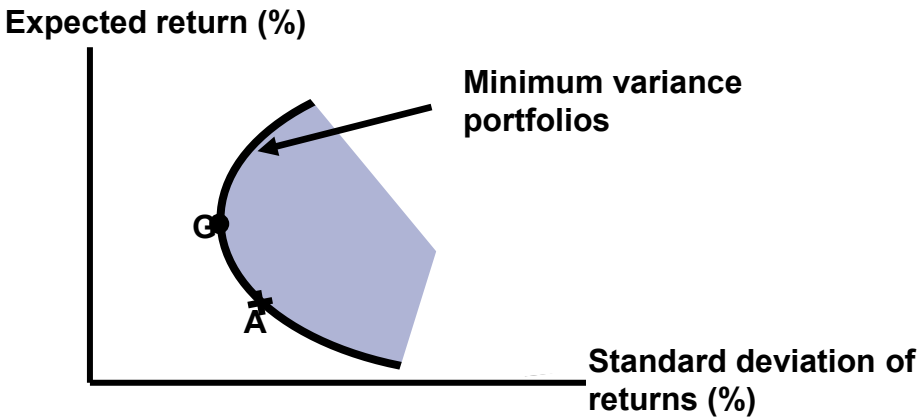
Depicts the predicted plot of the set of feasible portfolios constructed from all stocks listed on the ASX

Choosing the optimum portfolio

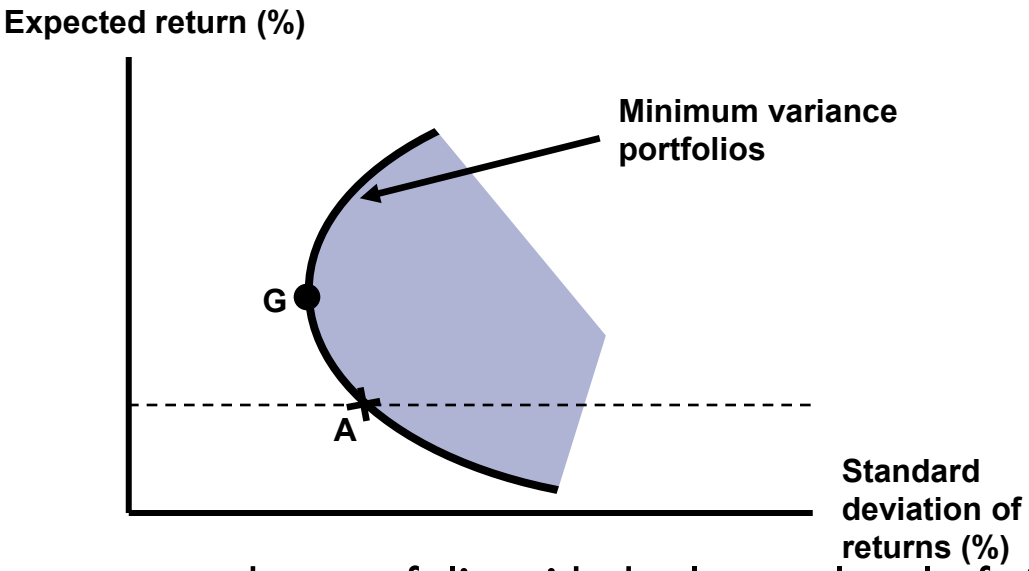
Of the feasible portfolios, which portfolio is optimum or 'rational'?

Minimum variance portfolios

- The subset of feasible portfolios that offer the lowest level of risk for each level of return



Choosing the optimum portfolio (cont)

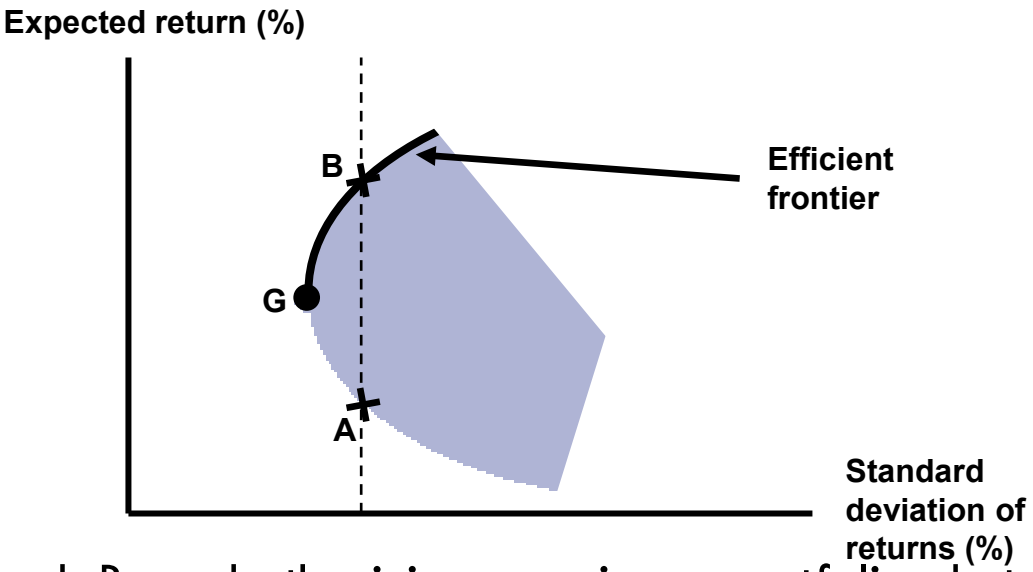


- Portfolio A represents the portfolio with the lowest level of risk given the level of return marked by the dotted line
- Portfolio G is the global minimum variance portfolio

Choosing the optimum portfolio (cont)

- Professor Harry Markowitz (1952) extended this analysis to identify the ***efficient set*** – a set of portfolios that a rational investor would choose
- This is a subset of the minimum variance portfolios offering the highest return for each level of risk
- This is effectively the top half of the set of minimum variance portfolios
- This set of portfolios is sometimes referred to as the ***efficient frontier***

Choosing the optimum portfolio (cont)

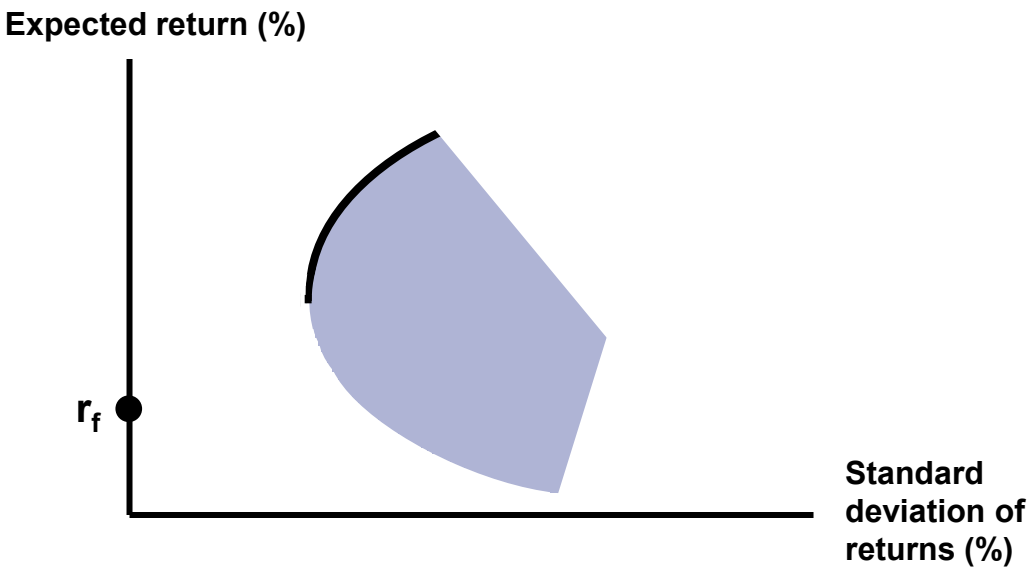


- Points A and B are both minimum variance portfolios, but B has a higher return than A for the same level of risk
- Rational investors would choose portfolio B over portfolio A

Choosing the optimum portfolio (cont)

- Based on this analysis, a rational investor will only choose portfolios on the efficient frontier – but which portfolio?
- The portfolio chosen will depend on each individual investor's attitude toward risk:
  - Those who are more risk-averse will choose portfolios close to point G – these portfolios have less risk and less return
  - Those who are less risk-averse will choose portfolios further to the right on the efficient frontier – these portfolios offer higher expected returns but carry greater risk

Optimum portfolios – multiple shares plus a risk-free asset

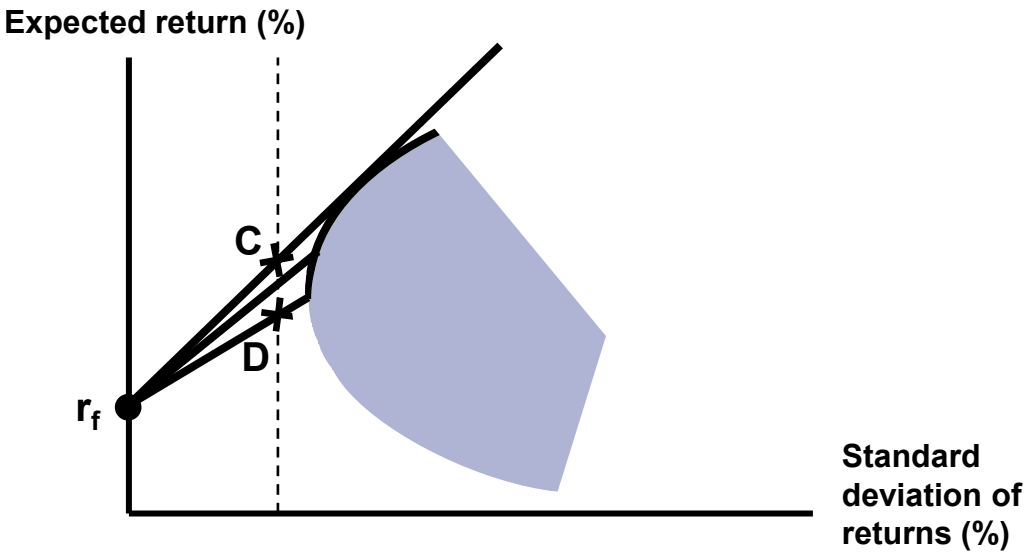


- Markowitz's analysis continues by assuming the existence of a risk-free asset, located on the y-axis as shown above because it's risk (measured by standard deviation) is zero

Optimum portfolios – multiple shares plus a risk-free asset (cont)

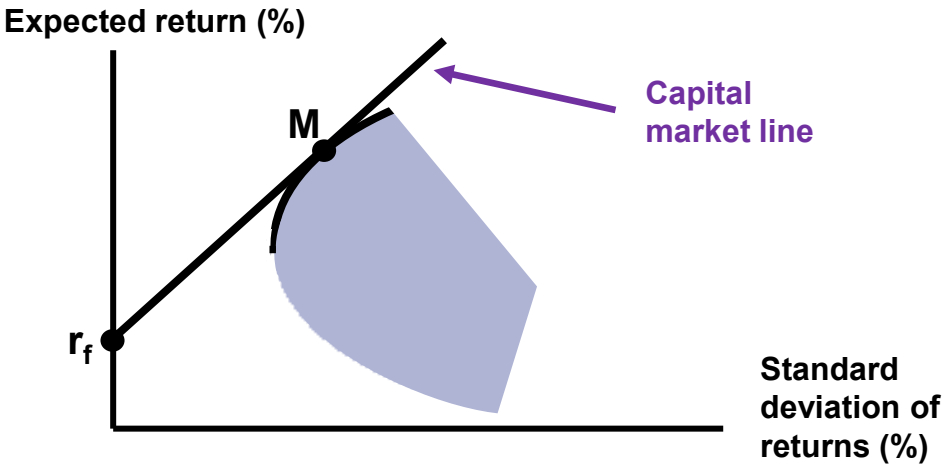
- New sets of feasible portfolios now exist
- These portfolios combine the risk-free asset and portfolios in the efficient set
- Each of the new sets of feasible portfolios is represented by a straight line from the risk-free asset to a portfolio on the efficient frontier
- The position of each portfolio on these lines depends on the weighting of the risk-free asset and the risky asset in the portfolio

Optimum portfolios – multiple stocks plus a risk-free asset (cont)



– Portfolios C and D are both now feasible, but C has a higher return than D, for the same level of risk, and hence is preferred

Optimum portfolios – multiple stocks plus a risk-free asset (cont)



- The line is called the *capital market line* (CML), and the point of tangency is represented by the portfolio M
- M is a theoretical portfolio containing all available stocks
- It is the *market portfolio*, and hence its level of risk and return are equal to those of the market overall

Optimal Portfolios II: CML

- The equation for the CML gives us an expression for the expected return on any efficient portfolio (i.e. a portfolio that lies on the CML):

$$r_p = r_f + \frac{r_m - r_f}{\sigma_m}\sigma_p$$

where:

r_p, r_f, r_m = return on the portfolio, the risk-free asset and the market, respectively

σ_p, σ_m = standard deviation of the portfolio and the market, respectively

Optimal Portfolios II: CML

- The formula for the Sharpe Ratio is:

$$Sharpe\ Ratio_p = \frac{ER_p - r_f}{\sigma_p}$$

- ER_p = Expected return of portfolio
- r_f = Risk-free rate
- σ_p = Standard dev of portfolio

- The higher the Sharpe Ratio the better the portfolio.
- The slope of the capital market line (CML) is the Sharpe Ratio.
- The market portfolio M has the highest possible Sharpe Ratio of any portfolio on the efficient frontier.

The End

Next week we will look at a model for determining the required return on a risky asset